

QUESTION

A.

DERPARTMENT OF METHEMATICS AND ACTUARIAL SCIENCE

END OF FISRT SEMESTER EXAMINATION

COURSE NAME: COMPUTATIONAL STATISTICS

COURSE CODE: ASTA 607

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A. Description of mass data

Description

The crabs data frame has 200 rows and 8 columns, describing 5 morphological measurements on 50 crabs each of two colour forms and both sexes, of the species *Leptograpsus variegatus* collected at Fremantle, W. Australia.

Usage

crabs

Source

Campbell, N.A. and Mahon, R.J. (1974) A multivariate study of variation in two species of rock crab of genus *Leptograpsus*. *Australian Journal of Zoology* **22**, 417–425.

B. Data Structure

```
> head(crabs)
  sp sex index  FL  RW  CL  CW  BD
1  B   M     1  8.1 6.7 16.1 19.0 7.0
2  B   M     2  8.8 7.7 18.1 20.8 7.4
3  B   M     3  9.2 7.8 19.0 22.4 7.7
4  B   M     4  9.6 7.9 20.1 23.1 8.2
5  B   M     5  9.8 8.0 20.3 23.0 8.2
6  B   M     6 10.8 9.0 23.0 26.5 9.8

>
```

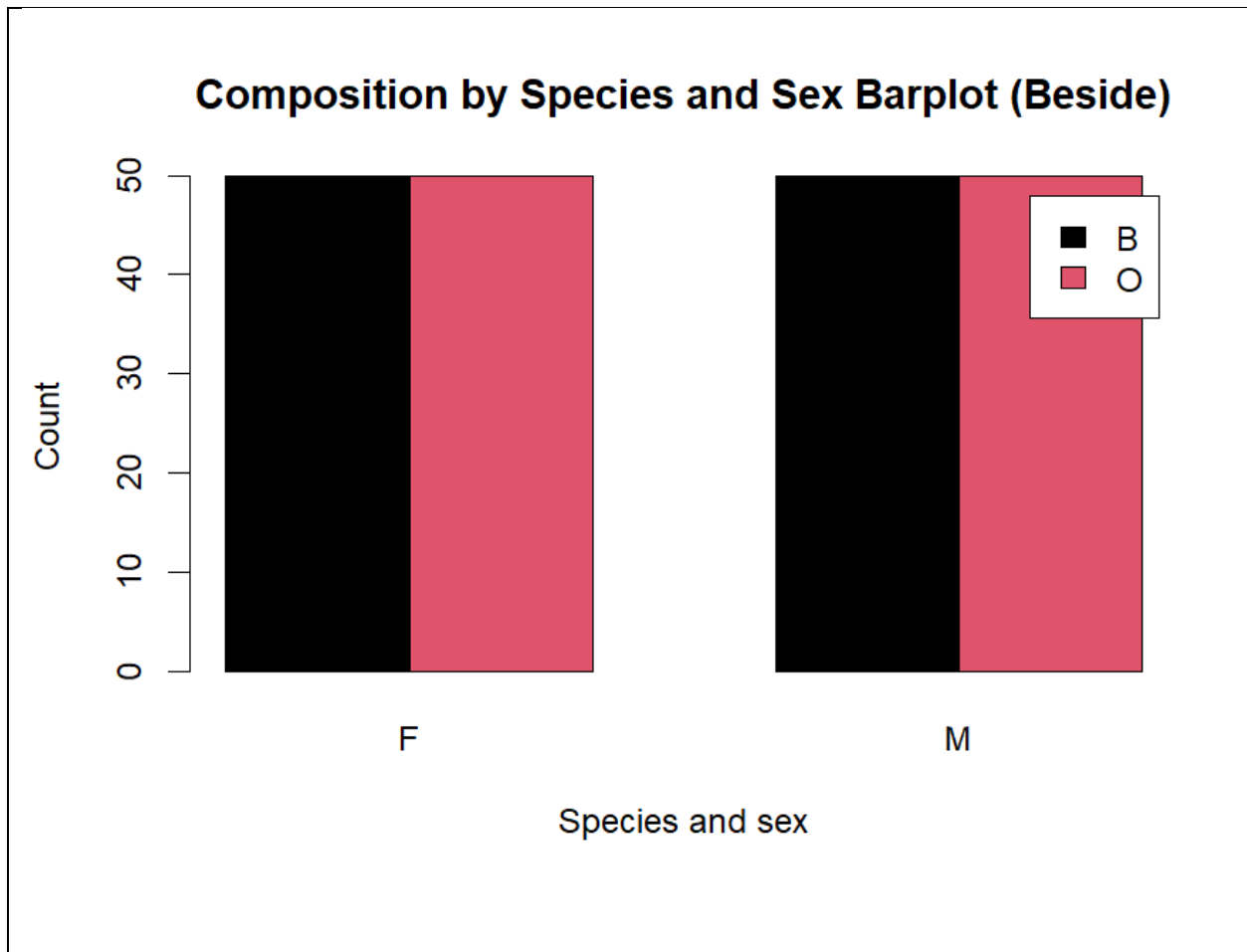
QUESTION

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C. Frequency table for species and sex

```
> freq_tab_species_sex <- table(Species = crabs$sp, Sex =  
crabs$sex)  
> freq_tab_species_sex  
      Sex  
Species F  M  
    B  50  50  
    O  50  50
```

D. Barplots for sex and species



QUESTION

A.

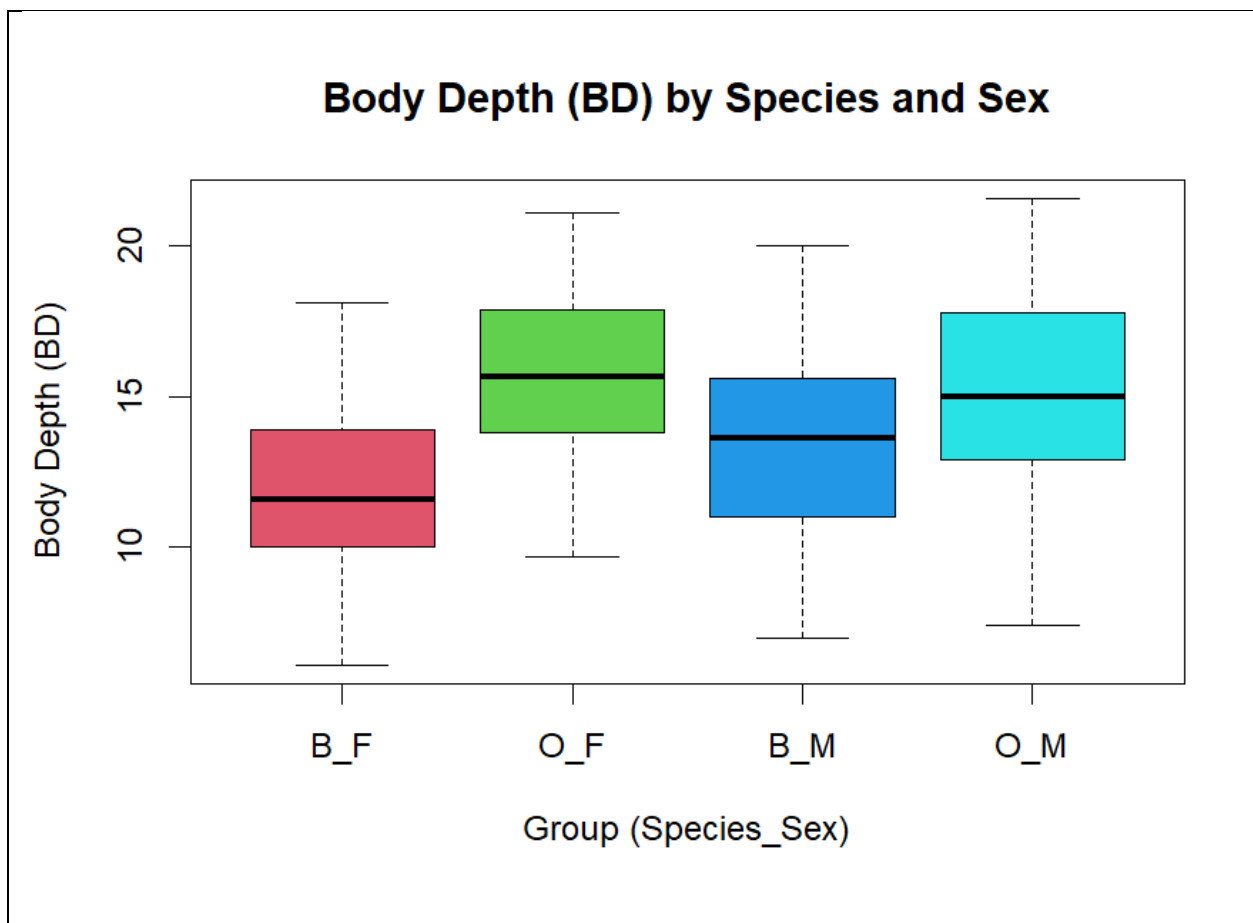
E.Comment on data set.

```
group_counts <- as.vector(tab_species_sex)
> min(group_counts)
[1] 50
> max(group_counts)
[1] 50
> min(group_counts) == max(group_counts)
[1] TRUE
```

Comment:from the above information and the equal frequency observed on the bar chart ,the data set is balanced across the groups

Question 2.

A.Side by side box plot.

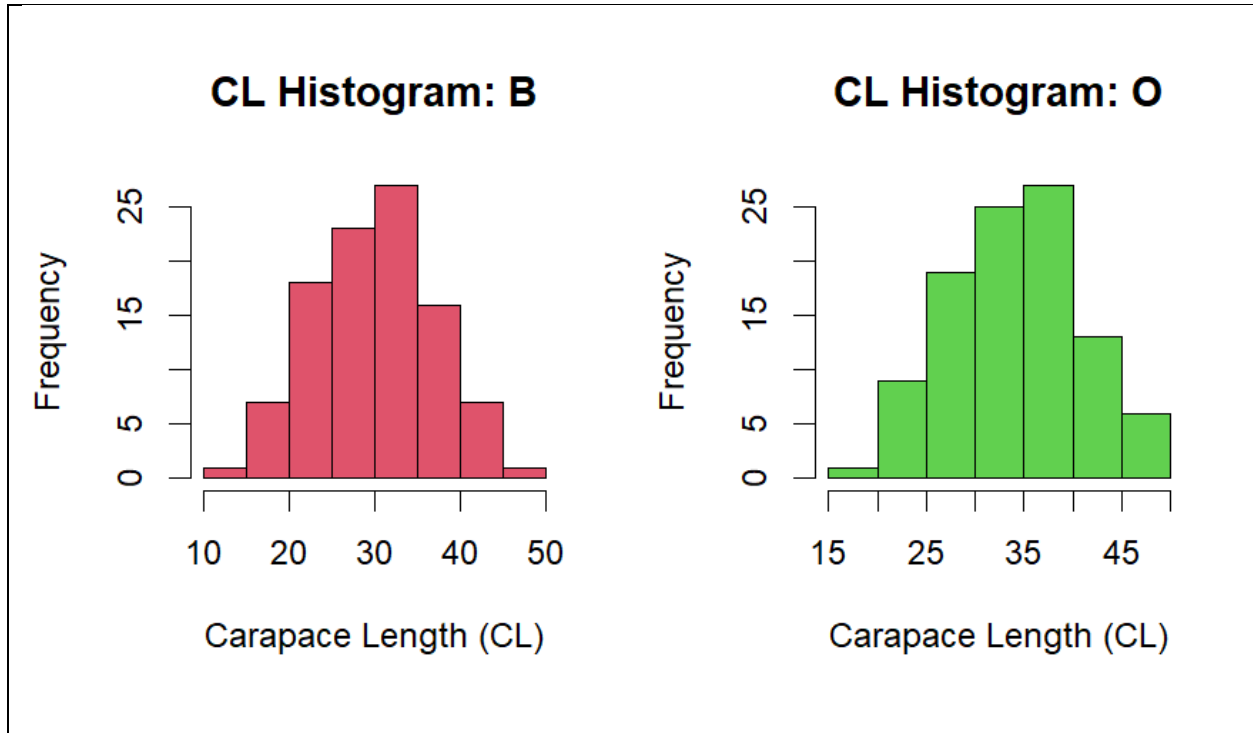


QUESTION

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Question 3.

A.



B. Skewness and Modality

```
# (b) Skewness and modality support (numeric summaries)
> by(crabs$CL, crabs$sp, summary)
crabs$sp: B
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 14.70  24.85   30.10   30.06  34.60   47.10
-----
crabs$sp: O
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 16.70  29.20   34.55   34.15  39.40   47.60
```

Comment:

- For both species, the **mean and median are very close**.
- This suggests the distributions are approximately **symmetrical** (little or no skewness).
- Species **O** has **larger carapace lengths on average** than Species B.

QUESTION

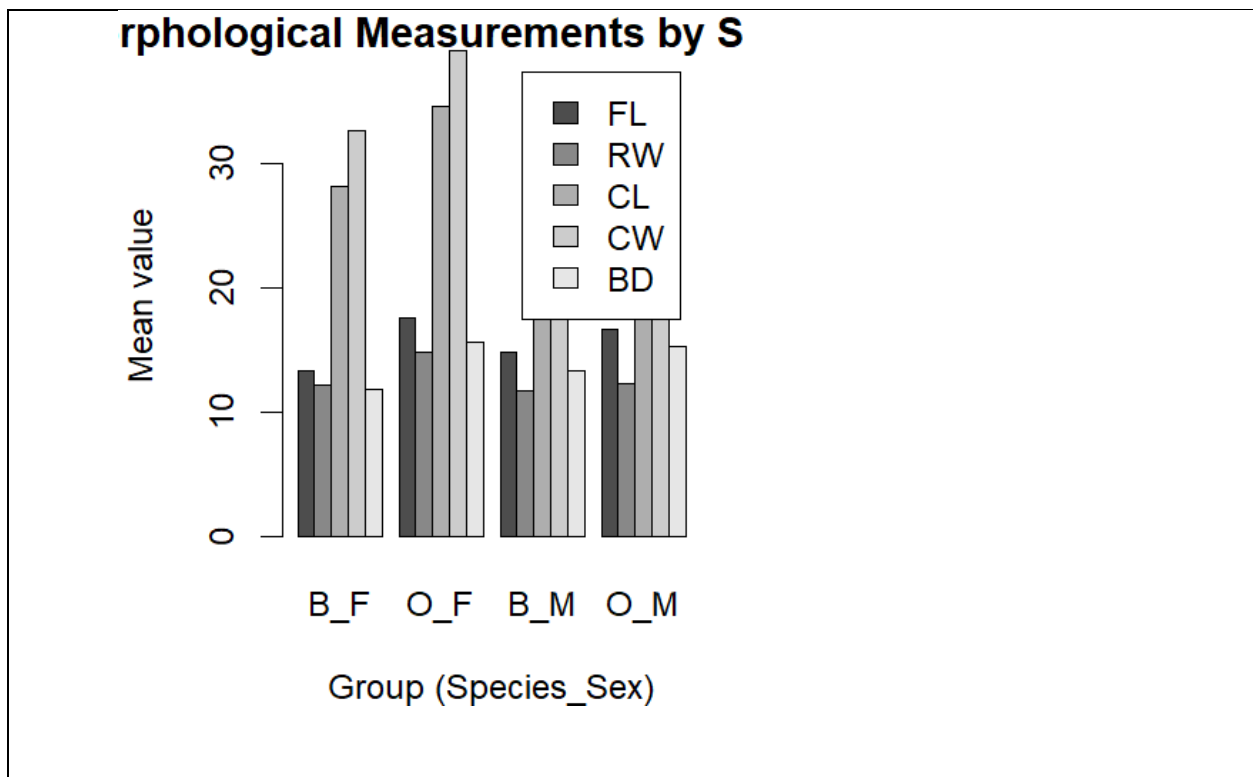
A.

So, Species O crabs tend to be bigger than Species B crabs.

Question 4.

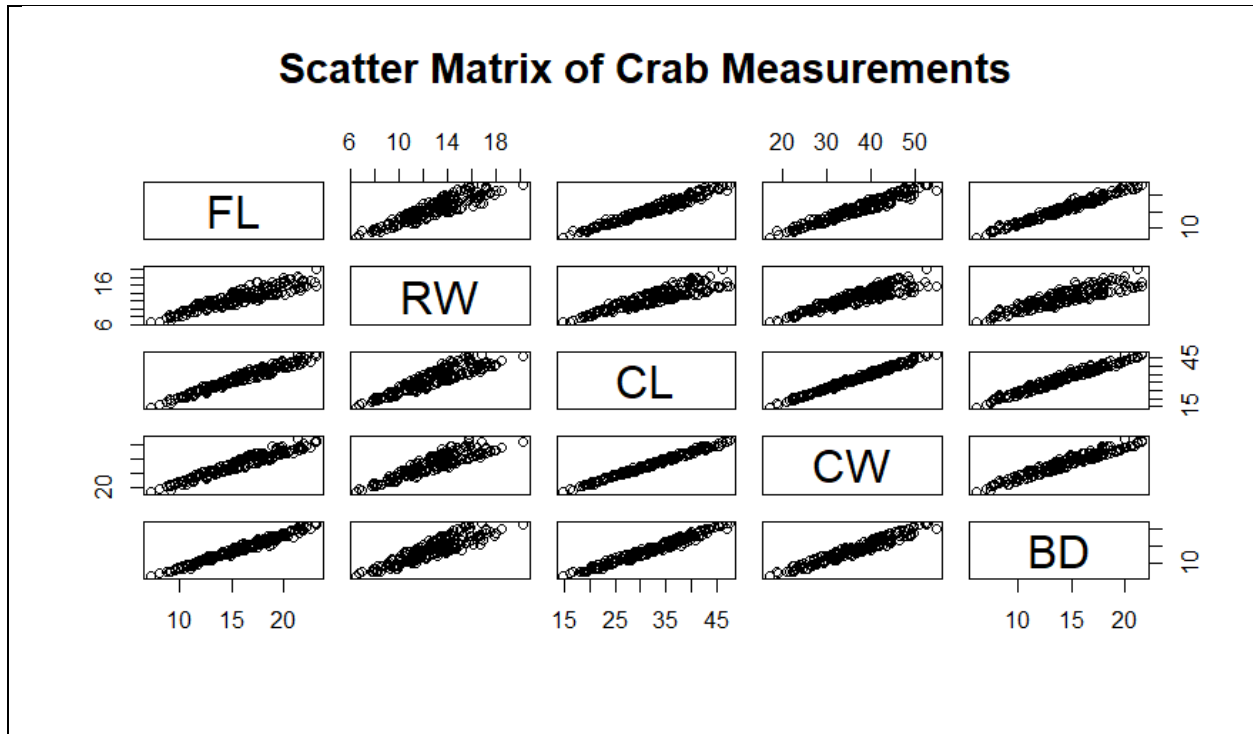
```
# (a) Mean of each morphological measurement (overall)
> overall_means <- colMeans(crabs[, c("FL","RW","CL","CW","BD")])
> overall_means
      FL      RW      CL      CW      BD
15.5830 12.7385 32.1055 36.4145 14.0305
>
```

B. Group Bar chat.



QUESTION

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B.

```
# (b) Strength/nature of relationships: correlation matrix
> cor(crabs[, c("FL","RW","CL","CW","BD")])
      FL      RW      CL      CW      BD
FL 1.000000 0.9069876 0.9788418 0.9649558 0.9876272
RW 0.9069876 1.0000000 0.8927430 0.9004021 0.8892054
CL 0.9788418 0.8927430 1.0000000 0.9950225 0.9832038
CW 0.9649558 0.9004021 0.9950225 1.0000000 0.9678117
BD 0.9876272 0.8892054 0.9832038 0.9678117 1.0000000
```

(e) COMMENT